St. Xavier's College (Autonomous), Ahmedabad-380009

M.Sc. Physics Syllabus

Semester-1

CORE Paper: Mathematical Physics-1 & Quantum Mechanics -1
Course Code: PPH-1801
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Use of Fourier transforms and its inverse in practical applications of Physics, apply Laplace transform and its inverse to solve initial value and other related problems.
2. Demonstrate good understanding of basic concepts in group theory, Demonstrate good understanding of various types tensor and will develop basic skills to do tensor algebra
3. Decide whether a given group is cyclic, and given a finite cyclic group, find a generator for a subgroup of a given order, Express a given finite cyclic group as the direct product of cyclic groups of prime power order and, given two direct products of cyclic groups, determine whether or not they are isomorphic
4. Express products of elements of a group defined by generators and relations in appropriate standard form, Understand concept of tensor variables and difference from scalar or vector variables.
5. Understand the reason why the tensor analysis is used and explain usefulness of the tensor analysis,
6. Clarify the basic concepts of perturbation and describe nondegenerate energy levels.
7. Quantum mechanical axioms and the matrix representation of quantum mechanics, apply the variational method and time-independent perturbation theory to solve simple problems
8. Will be able to solve the numericals of the above mentioned concepts and theory

Course Structure

Unit-1

Integral transform: Integral transform: Introduction, Development of Fourier Integral, Fourier Transforms-Inversion Theorem, Fourier Transforms of derivatives, Laplace transforms, Laplace transform of Derivatives, Properties: Substitution and Translation, Derivative of Transform, Convolution theorem, Inverse Laplace Transform

Unit-2

Group theory: Group, subgroups and classes, Invariant sub groups, factor groups, Homomorphism and Isomorphism, Group representation, Reducible and irreducible representation, Schur's Lemmas, orthogonality theorem, Character of a representation, Character tables, Decomposing a reducible representations into irreducible ones, Construction of representation, Representations of groups and quantum mechanics.
Tensor: Introduction, n-dimensional space, superscripts and subscripts, Coordinate transformations, Indicial summation conventions, Dummy and Real indices, Kronekar delta symbol, Scalars, Contra variant vectors and covariant vectors, Tensors of higher ranks, Algebraic operations, Symmetric and Ant symmetric tensors, Invariant tensors, Conjugate and reciprocal tensors, Relative and absolute tensors, Line element and matrix tensor, Fundamental tensors.

Unit-3

Time-Independent Perturbation Theory : Basic Concepts, Non degenerate Energy levels, Anharmonic oscillator: first order correction, the ground state of Helium, Effect of Electric field on the ground state of Hydrogen, Degenerate Energy levels, Effect of Electric field on the n=2 state of hydrogen, Spin orbit Interaction.
The variation method: The Variation Principal, Rayleigh-Ritz Method, Variation method for excited states. The Hellmann –Feynman Theorem, The ground state of Helium, The ground state of the Deuteron.

Unit-4

Heisenberg Method: The Heisenberg Method, Matrix Representation of Wave function, Matrix Representation of Operator, Properties of Matrix element, Schrodinger Equation in Matrix Form, Eigen value Problems, Unitary Transformations, Linear Harmonic Oscillator: Matrix method
WKB Approximation: The WKB Method, The Connection Formulas, Validity of WKB Method, Barrier Penetration, Alpha Emission, Bound States in a Potential Well

Reference Books

1. Boas M.L., Mathematical methods in the physical sciences, JW, 1966
2. Chattopadhyaya P.K., Mathematical Physics, Wiley Eastern Ltd.
3. Arfken G., Mathematical methods for Physicists, Academic Press, 1970

4. Elements of Group Theory: By A W Joshi, New Age Int. (2008)
5. Matrices and Tensors in Physics, A W Joshi Third Edition New Age Int. (2005)
6. Modern Quantum Mechanics, J. J. Sakurai & Jim Napoliitano, CUP
7. Principles of Quantum Mechanic, R. Shankar, Springer

St. Xavier's College (Autonomous), Ahmedabad-380009

M.Sc. Physics Syllabus

Semester-1

CORE Paper: Analog Electronics
Course Code: PPH-1802
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Learn the characteristics of an ideal OPAMP and practical OPAMP.
2. Do analysis of inverting and non-inverting amplifier including different applications of OPAMP through numerical
3. Design and analyse the integrator and differentiator circuit and will solve the linear equation through OPAMP circuits
4. EDescribe function of each components in Transformer coupled Amplifier, class- A, B, and AB push pull power amplifier.
5. Design the Transformer coupled amplifier, class- A, B, and AB push pull power amplifier and also design Multivibrators
6. Describe the basic principles of complimentary push pull amplifier.
7. Understand the working of 3 terminal and 4 terminal voltage regulator circuits
8. Learn the basics and application of some of the Thyristor family devices
9. Learn depletion and enhancement type of MOSFET, Including the difference between JFET and MOSFET.
10. Understand and learn TTL logic family, CMOS, Gates and interfaces among TTL and CMOS.

Course Structure

Unit-1

Operational Amplifier and its Application: DC characteristics, input bias current, input offset current, input offset voltage, total output offset voltage, thermal drift, slew rate Basic OPAMP application, inverting and non inverting amplifier, scale changer, inverting summing amplifier, non inverting summing amplifier, subtractor, Adder- subtractor, Instrumentation amplifier, V to I and I to V converter, half wave rectifier and fullwave rectifier, clipper, clamper, Log and Antilog amplifier, Integrator and differentiator circuit, Electronics analog computation, comparator

Unit-2

Class-A Power Amplifiers: Class-A Power Amplifiers: Class-A direct coupled resistive load, Transformer Coupled Resistive Load, Design Theory, Power Amplifier Design, Harmonic Distortion , Power Output, Variation of Output power with load, Output Transformer Saturation, Disadvantages of a single ended transformer coupled amplifier, Push-pull Amplifier, Description of operation of class-A push-pull Amplifier, Theory of operation of class-A push-pull Amplifier.
Class-B Power Amplifier: The class-B push-pull Amplifier, Cross Over Distortion, Class-AB push-pull Amplifier, Transistor Phase Inverter, Conversion Efficiency of Class-B Amplifier, Relation between maximum output power and load, Relation between maximum output power and Transistor dissipation, Design of class-B push-pull amplifier, Other class-B push-pull amplifiers, Complementary Symmetry, Practical complementary symmetry amplifier.
Multivibrators: Astable multivibrator, Monostable muitivibrator, Bistable multivibrator, Schmitt trigger (Transistorised circuit)

Unit-3

IC voltage regulator : Internal circuit arrangement, Zener reference regulation protection , error amp, series pass transistor,3 terminal positive voltage regulators,3 terminal negative voltage regulators, 3 and four terminal adjustable voltage regulators, 4 terminal positive voltage regulators, 4 terminal negative voltage regulators, dual non tracking voltage reg, dual tracking voltage regulator.
Thyristor : SCR, Working of SCR, equivalent circuit of SCR, Important terms, V- I characteristics of SCR,SCR in normal operation, SCR H.W. rectifier, SCR F.W. rectifier, Applications of SCR, TRIAC, TRIAC construction, TRIAC operation, TRIAC characteristic, Application of TRIAC

Unit-4

MOSFET: Depletion MOSFET, Enhancement MOSFET, Difference between JFET and MOSFET, Handling precaution for MOSFET, Dual gate MOSFET, Integral gate protection
Logic family: Switching circuit, 7400TTL, TTL parameters, TTL overview, OPEN collector gates, Three state TTL device, External drive for TTL loads, 74C00 CMOS, CMOS inverter, NOR and NAND gates, CMOS characteristics, TTL-TO-CMOS interface, CMOS-TO- TTL interface, Emitter coupled logic(ECL)

Reference Books

1. Linear Integrated circuit 4th edition, By Roy Choudhury and Jain, New Age International
2. Integrated Electronics by MillmanHalkias
3. Operational Amplifier by Gayakwad
4. Operational Amplifier and Linear Integrated circuit By Coughlin and Driscoll, Pearson
5. Electronic Devices and Circuit by Allen Mottershead
6. Electronics Devices and Circuits by S. Shalivahanan, N.SureshKumar,TMH.
7. Hand book of Electronics by Kumar and Gupta
8. Electronics Principle by Malvino Bates, TMH

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M.Sc. Physics Syllabus

Semester-1

CORE Paper: Numerical methods and Computer programming in C
Course Code: PPH-1803
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Apply numerical methods to obtain approximate solutions to mathematical problems.
2. Derive numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations.
3. Understanding of Monte-Carlo evaluation of integrals and error and apply in numerical methods.
4. Analyse and evaluate the accuracy of common numerical methods.
5. Implement numerical methods in C Language
6. Be able to perform Monte-Carlo simulations of simple systems,
7. Able to develop C programs.
8. Ability to design and develop Computer programs, analyse, and interprets the concept of pointers, declarations, initialization, operations on pointers and their usage.
9. Able to define data types and use them in simple data processing applications also he/she must be able to use the concept of array of structures.
10. Develop confidence for self-education and ability for life-long learning needed for Computer language.

Course Structure

Unit-1

Numerical Methods: Solution of algebraic and transcendental equations: Iterative, bisection and Newton-Raphson methods, Solution of simultaneous linear equations: Matrix inversion method, Interpolation: Newton and Lagrange formulas, Numerical differentiation, Numerical Integration, Trapezoidal, Simpson and Gaussian quadrature methods, Leastsquare curve fitting, Straight line and polynomial fits, Numerical solution of ordinary differential equations: Euler and Runge-Kutta methods

Unit-2

Simulation: Generation of uniformly distributed random integers, Statistical tests of randomness, Monte-Carlo evaluation of integrals and error analysis, Non-uniform probability distributions, Importance sampling, Rejection method, Metropolis algorithm, Molecular diffusion and Brownian motion as random walk problems and their MonteCarlo simulation

Unit-3

C Language Programming: Basic structure of a C program, Executing a C program, Keywords and identifiers, Variables, C operators, Reading a character, Writing a character, Formatted input, Formatted output, Decision making with if statements, simple if statement, if-else statement, nesting of if-else statements, else if ladder, switch statement, conditional operator, go to statement, While statement, do statement, do while, for statement, jumps in loops – continue and break statements. One dimensional arrays, declaration and initialization of arrays, two dimensional and multi dimensional array

Unit-4

C Language Programming: Declaring and initializing string variables, reading and writing strings, arithmetic operations on characters, concatenation, comparing, copying and finding length of strings, string handling functions, table of strings. Need for user defined functions, the form of C functions, return values and their types, calling a function, category of functions, noninteger functions, nesting of functions, recursion, functions with arrays, scope visibility and lifetime of variables, ANSI C functions

Reference Books

1. Computer Oriented Numerical Methods: Rajaraman
2. Computational Methods in Physics and Engineering: Wong
3. A Guide to Monte Carlo Simulations in Statistical Physics: Landau and Binder
4. Applied Numerical Analysis: Gerald
5. Balagurusamy E: Programming in ANSI-C (IInd.Ed.)TMHpub.

6. P.Day and M.Ghosh. Programming in C, Oxford University press,2007

7. Gottfried B.S. Programming with C

8. Kenetker Y. Let us C, BPB pub.

9. Kernighan B.W and Ritchie D.K C programming language, PH pub.

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M.Sc. Physics Syllabus

Semester-1

CORE Paper: Solid State Physics
Course Code: PPH-1804
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Understand Bloch's theorem and what energy bands are and know the fundamental principles of semiconductors
2. Explain the formation of energy band and classify metals, semiconductors and insulators, understand different models and methods like the nearly free electron model, Tight binding approximation, Wigner-Seitz cellular method and Pseudopotential method.
3. Describe Fermi Surface, its construction and will be able to apply it to study the heat capacity, electrical conductivity and thermal conductivity of the material.
4. Distinguish monovalent, divalent, trivalent and tetravalent Fermi surfaces and its characteristics, Describe the electric and magnetic field on Fermi surfaces.
5. A detailed discussion on ferromagnetism and anti-ferromagnetism that make students enable to describe few models like Weiss molecular theory for ferro, hysteresis, and classifications of soft and hard material on that basis.
6. Students can also explain Heisenberg spin and exchange interaction in ferromagnetism super exchange in anti-ferromagnetism
7. Define superconductivity and give a qualitative description of all properties including Meissner effect
8. Show how the London equations and Maxwell's equations lead to the prediction of the Meissner effect.

9. Apply London theory and Ginzburg-Landau theory for superconductivity for type-I and type-II superconductivity based on thermodynamic calculations of the Gibbs free energy for a superconductor

10. Discuss the fundamental theories of superconductivity and their application

Course Structure

Unit-1

Band theory of solids : Review: Developments of energy bands in solids, Electron motion in crystal (one electron approach), Bloch theorem, Kronig-Penny model, Numbers of possible waves functions in a band, Crystal momentum, The concept of effective mass, Concept of holes, Metals, Insulators and semiconductors, From isolated atoms to the freeelectron limit, The nearly free electron model, Tight binding approximation, Wigner-Seitz cellular method, Orthogonalised Plane Wave (OPW) method, Pseudopotential Method.

Unit-2

Fermi Surfaces: Free electrons, Square Lattice, Variations in the free electron Fermi surface of a square lattice according to valence of the metal, Harisson's construction of Fermi surface for square Bravais lattice (nearly free electrons), Fermi surface Brillouin zone, Divalent metals, Constant energy contours, Fermi surface three dimension lattices, Characteristics of Fermi surface, Effect of electric field on Fermi surface, Closed orbits and open orbits, Topology of orbits in magnetic fields, Experimental determination of the Fermi surface, DE Hass-Wan Alphen effect, Cyclotron resonance in metals.

Unit-3

Magnetic Properties: Ferromagnetism and Antiferromagnetism: Review: Diamagnetism and Paramagnetism. Ferromagnetism (Weiss molecular field theory), A two electron system (Singlet and Triplet) states, Calculation of Singlet Triplet splitting (exchange-splitting), Heisenberg spin Hamiltonian, Heisenberg's exchange interactions in ferromagnets, Ferromagnetism in Iron, Cobalt and Nickel (non-integral values of magnetization), Ferromagnetics domains and Hysteresis, Closure domains, The Bloch wall, Antiferromagnetism, Ferrimagnetism, Direct exchange and super exchange, Super exchange in antiferromagnets, Spin Waves and Magnons, Magnons in a one-dimensional Ferromagnet, Soft and Hard magnetic materials.

Unit-4

Superconductivity: Review: The discovery,Occurrence of superconductivity, The high temperature superconductors, Structure of high-Tc superconductors, Isotope effect, Critical magnetic field, Meissner effect Specific heat, Density of state and energy gape, Thermodynamics of a superconductor, The London theory, Pippard's equation and coherence length, Variation of order parameter across a 'superconductor-normal interface', Type I and II superconductors, The mixed

state, GinzbergLandau theory, The Bardeen-Cooper-Schrieffer (BCS) theory of superconductivity, The BCS ground state, Flux quantization, Josephson junction and Josephson effect, SQUIDs, Properties of HighTemperature Superconductors (HTSCs), Applications of superconductivity.

Reference Books

1. Solid State Physics, Ajay Kumar Saxena, Macmillan India Limited
2. Essentials of Solid State Physics, S P Kuila, New Central book agency (P) Ltd.
3. Solid State Physics, S O Pillai, New Age International (P) Ltd.
4. Introduction to Solid State Physics, Charles Kittel, Wiley India (P) Ltd, New Delhi
5. Elements of Solid State Physics, J. P. Srivastava, PHI, India.
6. Intermediate Quantum Theory of Crystalline Solids, A. O. E. animalu, Printice-Hall of India Pvt. Ltd.

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M.Sc. Physics Syllabus

Semester-1

CORE Paper: Electronics Laboratory
Course Code: PPH-1805L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to demonstrate the OPAMP parameters.
2. Will be able to demonstrate the OPAMP as inverting or non-inverting amplifiers.
3. Will be able to demonstrate the OPAMP as Voltage to Current and Current to Voltage Converter.
4. Will be able to demonstrate the OPAMP as an Integrator.
5. Will be able to demonstrate the OPAMP as a differentiator.
6. Will be able to demonstrate the Voltage regulator using OPAMP.
7. Will be able to demonstrate the OPAMP as an adder and subtractor.
8. Will be able to demonstrate 5V regulated power supply using three terminal IC
9. Will understand the application of OPAMP and practical use.
10. Will be ready to design the circuit using OPAMP characteristics.

Course Structure

1. Study of OPAMP parameters
2. OPAMP as a inverting/non-inverting amplifier

3. OPAMP as V to I and I to V converter
4. OPAMP as an integrator
5. OPAMP as a differentiator
6. Voltage regulator using OPAMP
7. OPAMP as an adder and subtractor
8. 5V regulated power supply using three terminal IC

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M.Sc. Physics Syllabus

Semester-1

CORE Paper: Physics Laboratory
Course Code: PPH-1806L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to measure the speed of the sound wave using CRO
2. Will be able to determine the specification rotation of sugar solution using Polari meter
3. Will be able to measure surface tension of liquid using surface wave
4. Will be able to measure dielectric constant of a non-polar liquid
5. Will be able to measure the energy band gap, voltage coefficient and diffusion capacitance of the PN junction diode.
6. Will be able to measure the resistivity using four probe.
7. Will be able to run scripts in —C || Programming.
8. Will be able to do numerical methods using —C || programming
9. Will understand the application and importance of —C || programming.
10. Will be able to explain importance of physics practical.

Course Structure

1. Measurement of the speed of the sound wave using CRO
2. To determine specification rotation of sugar solution using Polari meter

3. Measurement of surface tension of liquid using surface wave
4. Dielectric constant of a non polar liquid
5. Measurement of the energy band gap, voltage coefficient and diffusion capacitance of the PN junction diode
6. Measurement of the resistivity using four probe
7. Numerical methods using —C programming
8. Numerical methods using —C programming

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M.Sc. Physics Syllabus

Semester-2

CORE Paper: Digital Electronics and Microprocessor
Course Code: PPH-2801
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Understand 1-bit memory element, learn truth table, excitation table and FSM of different type of flip-flop.
2. Design/convert the flip-flop.
3. Learn application of flip flop as a shift register.
4. Understand working of SISO, SIPO, PIPO and PISO registers.
5. Get information about 555 timer IC.
6. Designing the different type of multivibrator using 555 IC.
7. Study different types of digital counter in detail. Also learn the synchronous and asynchronous counters
8. Study various types of digital to analog converters and analog to digital converters.
9. Study basic working and functioning of microprocessor 8085, with its organization and architecture.
10. Also perform some of the basic programming of microprocessor with additional instruction sets.

Course Structure

Unit-1

555- Timer: Description of functional diagram, Monostable operation, Astable operation, Bistable operation, Schmitt trigger circuit
Sequential logic circuits: Flip- Flops: 1- bit memory cell using transistor, 1- bit memory cell using NOR gates, clocked SR flip flop, clocked S-R flip flop with preset and clear inputs, J-K flip flop, Race around condition, Master slave flip flop, D- flip flop, T- flip flop, characteristics equation of flip flop, Flip flop as finite state machine, flip flop excitation table, Flip flop conversion (S-R to T , D and J-K flip flop, T to D flip flop, D to T flip flop)
Registers and Shift registers: Registers, SISO shift register, SIPO shift registers, PISO shift registers,PIPO shift registers, Bidirectional shift registers, Universal registers

Unit-2

Counters: Asynchronous counter, Ripple counter, Decoding gates, Synchronous counter, Mode -3 counter, Mode-5 counter, Mode-10 counter, Synchronous counter design, Design of UP- down counter, Analysis of clocked sequential circuit, State table, state diagram, Design of state sequential circuit
A/D and D/A converter: Basic principle of DAC, DAC circuits, Resister divider DAC, R/2R ladder network DAC, DAC specification, Basic principle of ADC, ADC circuits, Parallel comparator ADC, Counter type ADC, Successive approximation type ADC, Dual slope ADC, Specification of ADC

Unit-3

Microprocessors and its Architecture: Microprocessors, Microprocessor as a CPU, Organization of a Microprocessor based system, working of a Microprocessor, Languages employed in a Microprocessor, 8085 MPU and its architecture, Microprocessor Communication and Bus Timings.

Unit-4

Microprocessor Programming: Introduction to 8085 MPU Instructions, 8085 MPU programming with additional instructions.

Reference Books

1. Linear Integrated circuit 4th edition, By Roy Choudhury and Jain, New Age International
2. Digital Electronics By G K Kharate, Oxford university Press
3. Digital Principles and Applications ByMalvino, Leach and Saha
4. Digital Design By Morris Mano
5. Microprocessor Architecture, Programming, and Applications with the 8085, sixth edition, Ramesh Gaonkar, Penram International Publications.

6. Microprocessor Programing and Interfacing by N.K. Srinath; PHI publications.

7. Microprocessor and its interfacing by Sunil Mathur, PHI publications.

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M.Sc. Physics Syllabus

Semester-2

CORE Paper: Electrodynamics
Course Code: PPH-2802
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Understand different conservation laws.
2. Solve problems of electromagnetic waves and wave propagation.
3. Describe the nature of electromagnetic wave and its propagation through different media and interfaces.
4. This unit is aims to teach advance formulation of retarded scalar and vector potentials for generalized description of electromagnetic fields originating from moving and accelerating point charges.
5. Understand the concept of dipole radiation and its application to design and optimize the antenna properties.
6. Describe the concept of electromagnetic waves radiation for electric and magnetic dipole moment.
7. Understand the relativity concepts of electrodynamics, and discussion of Lienard Wiechert potential.
8. Understand and analyse the electrical quadruple and amount of total power radiation transmitted for different cases like arbitrary source and point charges.
9. This unit introduces students with concept of scattering of electromagnetic waves by dipoles, dielectric spheres and many other media with applications.
10. Understand the scattering and dispersion in detail and perturbation theory of scattering.
11. Solve the problems of the above mentioned concepts and theory

Course Structure

Unit-1

Conservation Laws: Charge and Energy,. The continuity equation, Poynnting's theorem, momentum, Newton's third law in electrodynamics, Maxwell Stress Tensor, Conservation of Momentum, Angular momentum, problems, Electromagnetic waves and wave propagation; Wave guides Absorption and Dispersion, Electromagnetic waves in conductors, Reflection at a Conducting surface, The frequency dependence of Permittivity, Guided waves, Wave guides, The TE waves in a rectangular Wave Guide, Resonant cavities, Cavity resonator - Faraday and Kerr effects. The Coaxial Transmission Line. Problems

Unit-2

Potential and Fields: The Potential formulation, Scalar and Vector Potential, Gauge Transformation, Coulomb Gauge and Lorentz Gauge, Continuous distributions, Retarded Potentials, Jefimenko Equations, Point charges, Lienard –Wiechert Potentials, The fields of a Moving Point Charge, Total power radiated by an accelerated charge: Larmour's formula and its relativistic generalization, Problems

Unit-3

Radiation: What is Radiation? Electric dipole radiation, Magnetic dipole radiation, Electric Quadrupole fields, Center fed Linear Antenna , Approximation of Sinusoidal current,, The Antenna as a Boundary value problem, Radiation from arbitrary source, Point charges, Power radiated by a point charges, radiation reaction, Abraham- Lorentz formula for radiation reaction, The Physical basis of radiation Reaction. Problems

Unit-4

Scattering and Dispersion: Scattering at long wavelengths, scattering by dipoles induced in a small scatterers, scattering by a small dielectric sphere, scattering by a perfectly conducting sphere, collection of scatterers, Perturbation theory of Scattering, Rayleigh's explanation of the blue sky, scattering by gas and liquids, attenuation in optical fiber: General theory, Born approximation, Blue sky, Density fluctuations: Critical Opalescence, Attenuation in optical fiber, Dispersion in Dilute gases, Dispersion in liquids and solids, Media containing free electrons, Problems

Reference Books

1. Griffiths, Introduction to electrodynamics, Prentice Hall India Ltd. (2nd ed.)
2. Classical Electrodynamics John David Jackson(Third Edition) Wiley Student edition.
3. Laud B. B., Electromagnetics, Wiley Eastern, (2nd ed.)

4. J.R. Reitz, F.J. Milford and R.W. Christy, 1986, Foundations of Electromagnetic Theory, 3rd Edition, Narosa Publication, New Delhi.

5. W. Panofsky and M. Phillips, 1962, Classical Electricity and Magnetism, Addison Wesley, London. 2.

6. J.D. Kraus and D.A. Fleisch, 1999, Electromagnetic with Applications, 5th Edition WCB McGraw-Hill, New York.

7. B. Chakraborty, 2002, Principles of Electrodynamics, Books and Allied, Kolkata.

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M.Sc. Physics Syllabus

Semester-2

CORE Paper: Classical Mechanics and Computaional Physics
Course Code: PPH-2803
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. To demonstrate knowledge and understanding of the fundamental concepts in Lagrangian and Hamiltonian formulation of mechanics
2. To represent the equations of motion for complicated mechanical systems using the Lagrangian and Hamiltonian formulation of classical mechanics.
3. To be familiar with the foundations of Theoretical Physics.
4. Understand Python programming language, solve physics problems using Python
5. Understand Mathematica Software
6. Solve physics problems using Mathematica Software

Course Structure

Unit-1

Canonical Transformation: An overview of Langrangian and Hamiltonian formulation, Noether's Theorem, Conservation Principles and Homogeneity of Space and Time. Canonical transformations: Generating functions, examples of canonical transformations, Poisson brackets and the Symplectic Condition, conservation theorem in Poisson bracket formalism, Jacobi's identity, angular momentum Poisson bracket relations.

Unit-2

Hamilton Jacobi Theory : Hamilton-Jacobi theory: Hamilton-Jacobi equations for principal and characteristic functions, Harmonic oscillator problem, Central force problem, Action-angle variables for systems with one-degree of freedom.
Small Oscillation: Eigen value equation, Free vibrations, Normal Coordinates, Vibrations of a triatomic molecule.

Unit-3

Numerical Methods using Python & Mathematica : Introduction to Python, Systems of Linear Algebraic Equations, Interpolation, Curve Fitting, Roots of Equations, Numerical Differentiation, Numerical Integration, Initial Value Problems, Matrix Algebra, Introduction to Mathematica and solving numerical methods.

Unit-4

Physics with Python & Mathematica : Classical Mechanics: Newtonian Mechanics, Central Forces, Calculus of Variation, Lagrange Dynamics, Hamiltonian Dynamics. Quantum Mechanics: The Schrodinger Equation, One- Dimensional Potential, The Harmonic Oscillator, Anharmonic Oscillator, Motion in the Central Force Field.

Reference Books

1. Classical Mechanics, H Goldstein, Pearson
2. Classical Mechanics, Rana & Joag, TMH
3. Mechanics, L D Landau E M Lifshitz, Elsevier Science Publications
4. Classical Mechanics, A K Raychaudhuri, Oxford University Press
5. Learning Python: Powerful Object-Oriented Programming, by Mark Lutz, O Reilly publication (5th Edition)
6. Numerical methods in Engineering with Python, by Jaan Kiusalaas, Cambridge University Press
7. The Mathematica Book by Stephen Wolfram, Cambridge University Press
8. Mathematica for Theoretical PhysicsVol - 1 and 2, Gred Baumann, Springer (2nd Edition)

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M.Sc. Physics Syllabus

Semester-2

CORE Paper: Quantum Mechanics-II, NMR, and LASERS
Course Code: PPH-2804
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Describe Symmetry transformation and conservation law, explain translation in space and time; deduce conservation of linear momentum and conservation of energy.
2. Discuss rotation in space and obtain conservation of angular momentum, identify space inversion: parity conservation.
3. Obtain eigen value of the square of the angular momentum operator and z component of angular momentum operator.
4. Derive the angular momentum matrices, state Pauli's spin matrix and derive spin matrix of a spin (1/2) system
5. Describe addition of angular momenta and obtain Clebsh-Gordan coefficient for a given system having two angular momenta
6. About fundamental of Nuclear Magnetic Resonance, measurement of NMR spectra and its interpretation, hardware, experimental methods, equivalent theoretical description and its applications like NMRI in daily life.
7. Describe the requirements for a system to act as a laser.
8. Differentiate the various types of lasers and their means of excitation, relate the structure and properties of lasers to their performance and intended applications

Course Structure

Unit-1

Symmetry and Conservation laws: Symmetry Transformation, Translation in Space: Conservation of Linear Momentum, Translation in Time: Conservation of Energy, Rotation in Space; Conservation of Angular momentum, Space Inversion; Parity Conservation, Time Reversal

Unit-2

Angular momentum : General Angular Momentum, Eigen value of J^2 and J_z , Angular Momentum matrices, Matrices for J_+ , J , J_x and J_y, Spin angular momentum, Spin vectors for Spin –(1/2) system, Addition of Angular momenta, Clebsh- Gordan Coefficient.

Unit-3

NMR and NQR Techniques: Theory of NMR - Bloch equations - Steady state solution of Bloch equations - Theory of chemical shifts - Experimental methods - Single Coil and double coil methods - Pulse Method - High resolution method - Applications of NMR to quantitative measurements. Quadruple Hamiltonian of NQR - Nuclear quadruple energy levels for axial and nonaxial symmetry - Experimental techniques and applications.

Unit-4

LASER: Requisites for producing LASER light, Basic principles , Einstein coefficients and Light Amplification, The threshold condition, Line broadening mechanism: Natural broadening, Collision broadening and Doppler broadening. Laser Rate Equation, Optical Resonators: Modes of a rectangular cavity, Spherical resonators, The Quality factor, Mode selection, Dye Laser , Excimer Laser.

Reference Books

1. Principles of Quantum Mechanics, R. Shankar, Springer
2. LASERS Theory and Applications – by K. Thyagarajan and A. K. Ghatak (Macmillan India Ltd., 2008).
3. Lasers and Non-linear Optics – by B. B. Laud (New Age International (P) Ltd., India, Second edition, 1996) .
4. Atom, Laser Spectroscopy - by S. N. Thakur D. K. Rai, PHI Learning Pvt. Ltd. New Delhi
5. LASER Fundamentals : Silfvast (Cambridge University , Press)
6. LASER's : Siegman (Univ. Science Books , USA)
7. Elements of Quantum Optics : Meystre and Sargent (Spriger – Verlag)
8. LASER Physics : Srgent, Scully and Lamb

9. Essentials of LASER and non – linear optics : Baruah Pragati Prakashan, Meerut)
10. Atta Ur Rahman, 1986, Nuclear Magnetic Resonance, Spinger Verlag, New York.
11. Raymond Chang, 1980, Basic Principles of Spectroscopy, Mc Graw-Hill, Kogakusha, Tokyo

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M.Sc. Physics Syllabus

Semester-2

CORE Paper: Physics Laboratory
Course Code: PPH-2805L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to demonstrate the Inverse square law and end point energy of beta particles G M tube.
2. CO2: Will be able to demonstrate the Ultrasonic Interferometer.
3. Will be able to demonstrate the Dipole Moment of organic molecule Acetone.
4. Will be able to demonstrate the Velocity of Liquid using Surface wave.
5. Will be able to demonstrate the Dielectric constant of a non-polar liquid
6. Will be able to demonstrate the Hall Effect.
7. Will be able to Measure the Lattice dynamics
8. Will be able to demonstrate Stefan's Constant of Radiation
9. Will understand the application of Ultrasonic Interferometer.
10. Will be ready to demonstrate application of dipole moment and dielectric constant

Course Structure

1. G M counter experiment(To verify Inverse square law and end point energy of beta particles G M tube)
2. Ultrasonic Interferometer

3. Dipole Moment of organic molecule Acetone
4. Velocity of Liquid using Surface wave
5. Dielectric constant of a non polar liquid
6. Hall Effect
7. Measurement of Lattice dynamics
8. Stefan's Constant of Radiation

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M.Sc. Physics Syllabus

Semester-2

CORE Paper: Electronics Laboratory
Course Code: PPH-2806L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to demonstrate the active filters using OPAMP.
2. Will be able to demonstrate the Wien Bridge Oscillator using OPAMP.
3. Will be able to demonstrate the Digital Design using Karnaugh Map.
4. Will be able to demonstrate the Digital Design using Multiplexer.
5. Will be able to demonstrate the conversion of Flip-Flop.
6. Will be able to demonstrate the decade asynchronous counter using 7490 IC.
7. Will be able to demonstrate the up-down synchronous counter using 74193 IC.
8. Will be able to demonstrate microprocessor programming.
9. Will understand the application of OPAMP and practical use.
10. Will be ready to design the circuit using OPAMP and flip-flop.

Course Structure

1. Active Filters Using OPAMP
2. Wien Bridge Oscillator Using OPAMP
3. Digital Design Using karnaugh map

4. Digital design using Multiplexer
5. Conversion of Flip-Flop
6. Decade Asynchronous Counter using 7490 IC
7. Up-Down Synchronous Counter using 74193 IC
8. Microprocessor programing

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M.Sc. Physics Syllabus

Semester-3

CORE Paper: Electronic Communication and Microprocessor Interfacing
Course Code: PPH-3801
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Understand the importance of the modulation in the communication
2. Get information about different types of the modulation.
3. Learn theory of Amplitude modulation and different techniques through numerical, learn theory of FM wave through numericals.
4. Get idea of generation of side band and bandwidth of the FM signal.
5. Learn practical circuit used for generation of AM and FM wave.
6. Learn the principle of FM demodulation and different FM detectors used in the FM receiver.
7. Learn the pulse modulation technique including pulse amplitude, pulse width, pulse position and pulse code modulation.
8. Also learn the digital modulation including ASK, FSK, and PSK.
9. Study the microprocessor 8085 with its advanced instruction sets including stack, subroutines, call and return instruction
10. Study the interfacing of microprocessor kit with peripheral devices such as DAC and 8255
11. Solve numerical based on above concepts and topics.

Course Structure

Unit-1

Amplitude modulation: Elements of analog communication, Amplitude modulation technique, Time domain representation of AM wave, power relation in AM wave, current relation in AM wave, modulation by several sine wave, Double sideband suppressed technique, single sideband technique, vestigial sideband modulation technique, Generation of AM signal. Angle modulation: Frequency modulation, phase modulation, comparison of frequency and phase modulation, frequency spectrum of FM wave, comparison of FM and AM wave, Generation of FM, Direct methods Basic FM demodulators, slope detection, Balanced slope detector, phase discriminator, ratio detector

Unit-2

Pulse Modulation Technique: Pulse amplitude modulation, pulse width modulation, pulse position modulation, pulse code modulation, Delta modulation, Differential code modulation, Demodulation of pulse digital modulated signal Digital Modulation Technique: Amplitude shift keying, Frequency shift keying, Phase shift keying Satellite communication: Satellite system, fixed satellite service, satellite communication Earth station, Antenna for satellite communication, advantages of cassegrain antenna, Antenna feed, Antenna structure.

Unit-3

Principle of light transmission in a fiber - propagation with in a fiber, fiber index profiles, Modes of propagation - overview of modes, key modal concepts, Maxwell's equations, waveguide equations for cylindrical fiber, wave equations for step index fiber, modal equation, modes in step index fiber, single mode fiber. Losses in fibers - absorption losses, leaky modes, mode coupling losses, bending losses, combined fiber losses. Dispersion - effect of dispersion on pulse transmission, intermodal dispersion, material (chromatic) dispersion, total dispersion and maximum transmission rates. Light sources and detectors for fiber optics. optical receiver circuit. Connectors and splices - losses in connectors and splices, connectors, fiber splices. Fiber optic communication link.

Unit-4

A: Stack; Subroutine; Restart, Conditional Call and Return Instructions; Advanced Subroutine Concepts.
B: The 8255A Programmable Peripheral Interface, Block Diagram of the 8255A; Mode 0: Simple Input or Output Port.
C: DAC Specifications; DAC 0800; DAC Interfacing

Reference Books

1. Kennedy's Electronic communication systems by Kennedy, Davis Prasanna(5th ed)
2. Electronics communication By Roddy and Coolen, 4th edition

3. Hand Book of Electronics by Kumar and Gupta

4. Electronics and Radio engineering By M L Gupta

5. Tosmasi W., Advanced Electronic Communication system, PHI.

6. Keiser G, Optical fiber communications, MGH, 2000

7. Senior J.M., Optical Fiber Communications – Principles and practice, Pearson, 2007

8. Fiber Optics and Optoelectronics, R. P. Khare

9. Microprocessor Architecture, Programming and Applications with 8085 5th edition by Ramesh Gaonkar for Units A and B:

10. 8 Bit Microprocessor—V.J. Vibhute and P.B. Borole

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M.Sc. Physics Syllabus

Semester-3

CORE Paper: Nano Science and Photo Voltaic System
Course Code: PPH-3802
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Learn some of the physical and chemical methods for the fabrication of Nanomaterial including various vacuum and non- vacuum deposition techniques
2. Understand some of the basic and advanced characterization techniques used to study the nanomaterial's as well as thin films.
3. Study mechanical, optical, structural, electrical and magnetic properties of nanomaterial. Further students will be studying Nano tubes of carbon including its types, preparation techniques, and some of the application.
4. At the end of the course students are able to describe the properties like physical, magnetic, conductivity etc. of nanomaterial's
5. Students are incapacity to explain the carbon nanotubes and quantum dots etc. and their applications.
6. Understand generation and separation of electron-hole pairs in a photovoltaic device.
7. As well as collection of charges, generation of photo-voltage and flow of current through photovoltaic device.
8. And study some of the solar cell characterizations, working configurations, solar cell structures and practical working problems of solar cells.
9. Understanding the solar simulator I-V measurements and quantum efficiency (QE) measurements.

10. Study of series and parallel connections of cell and mismatch in cell module.

11. Solve numerical based on above concepts and topics.

Course Structure

Unit-1

Synthesis of Nano materials- I (Physical Method): Introduction, Mechanical Methods, Methods based on Evaporation, Sputter Deposition, Chemical Vapour Deposition (CVD) , Electric Arc Deposition, Ion Beam Techniques (Ion Implantation), Molecular Beam Epitaxy (MBE)
Synthesis of Nano materials- II (Chemical Method): Introduction, Colloids and Colloids in Solutions, Growth of nano particles, Synthesis of metal nano particles by Colloidal Route, Synthesis of Semiconductor Nano particles by Colloidal Route, Langmuir-Blodgett(L-B) method , Micro emulsions, Sol gel Method
Characterization: Transmission electron microscopy (TEM), High Resolution TEM (HRTEM), Particle Size Analyzer, Scanning tunneling microscopy (STM), Atomic-force microscopy (AFM), Scanning near field optical microscopy (SNOM), Raman Spectroscopy and Photoluminescence.

Unit-2

Properties of Nano Materials: Introduction, Mechanical properties, Structural properties, Melting of Nano particles Electrical conductivity, Optical properties, Magnetic properties. Nano Tubes New form of carbon: Types of nanotubes - Formation of nanotubes - Various techniques - Preparation and properties of nanotubes - Uses of nanotubes and applications - Nano material processing for nanotube - Light and Nano technology - Nanoholes and photons - Quantum electronic devices - Quantum electronic devices - Quantum information and Quantum Computers.
Applications: Single electron tunneling, IR detectors, quantum dot lasers, targeted drug delivery using nanoparticles, nanosensors based on quantum size effects.

Unit-3

Theory of thin film nucleation –Impingement, adsorption and thermal accommodation, The capillarity model, Atomistic model, Four stages of thin film growth, cluster coalescence and depletion, thickness measurement by QCTM and Stylus profilometry methods. Lithography: photo, X-ray and e-beam.

Unit-4

P-N Junction under illumination: solar cell, Generation of photo voltage, Light generated current, I-V equation of solar cells, solar cell characteristics. Generation of carriers: rate of generation, absorption coefficient. Recombination of carriers: carrier life time, band to band recombination, Trap assisted recombination, Auger recombination. Losses in solar cell, effect of series and shunt resistance on efficiency. Analytical techniques: solar simulator I-V measurement, Quantum efficiency (QE) measurement, minority carrier life time and diffusion length measurement. Solar PV modules from solar cells: series and parallel connections of cell and mismatch in cell module.

Mismatch in series connection, Hot spots in the module, By pass diode, mismatching in parallel connection.

Reference Books

1. Nano-science and Technology by V.S. Muralidharan and A. Subramania (Ane Book Pvt. Ltd.).
2. Introduction to Nanotechnology by Charles P. Poole, and Frank J. Owens (Wiley India Pvt. Ltd
3. Nanotechnology: Principles and practice: SulbhaKulkarni.Capital Publishing Company
4. Nano: The Essentials –Understanding Nanoscience and Nanotechnology by T. Pradeep (McGrawHill Education).
5. Semiconductor material and Devices Characterization, By Dieter K. Schroder, John Wiley and Sons, NY1990. 1 Dr. Savan Patel
6. Integrated circuits by K. R. Botkar (Khanna Publishers).
7. Handbook of thin film technology by L.I. Maissel and R. Glang (McGraw-Hill).
8. The Materials Science of thin films by Milton Ohring (Academic press)
9. Thin Film Phenomena by K. L. Chopra (McGraw-Hill).
10. Solar Photovoltaics [Fundamentals, Technologies and Applications] by Chetan Singh Solanki (PHI Learning Private Limited, second edition 2013)
11. Electronic Devices and Components, by J Seymore (Longmann Scientific Technical)
12. Integrated Electronics, by K. R. Botkar, (Khanna Publishers.)
13. Integrated Electronics: Analog and Digital Circuits Systems, by J Millman and C. C. Halkias (Tata McGraw -Hill Publishing Company Ltd.)
14. Solid State Pulse Circuits, by David A. Bell (Prentice Hall of India Pvt. Ltd)
15. Energy Technology (Non-conventional, Renewable and conventional), by S. Rao and Dr. P. B. Parulkar (Khanna Publishers.)

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M.Sc. Physics Syllabus

Semester-3

CORE Paper: Adv. Topics in Quantum and Statistical Mechanics
Course Code: PPH-3803
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Have a working knowledge of differentiability for complex functions and be familiar with the Cauchy-Riemann equations
2. To evaluate integrals along a path in the complex plane and understand the statement of Cauchy's Theorem, use Green functions.
3. Clarify the basic concepts of scattering and illustrate Rutherford's scattering experiment. Describe scattering amplitude and scattering cross section and also apply partial wave analysis to find the same.
4. Establish the expansion of a plane waves in terms of infinite number of spherical waves and obtain Bauer's formula.
5. Give example of scattering by an attractive square well potential.
6. Outline the Born approximation and explain the validity of Born approximation and also develop skill to compute the phase shift for s and p wave scattering.
7. Will be able to demonstrate good understanding of fluctuations, its power spectrum and
8. will be able to apply it to study various types of noise.
9. Will be able to demonstrate good understanding of critical phenomena occurring at phase transition.
10. Have a basic understanding of the phase transitions,

11. Have a deep understanding of universality in second order phase transitions.
12. Will be familiar with order of phase transition and Clusius-Clayperon equation.
13. Understanding of Curie-Weiss theory of Magnetic transition.
14. Will be able to demonstrate Ising Model in zeroth approximation

Course Structure

Unit-1

Green's function method of solving inhomogeneous differential equations, Boundary Conditions, Application to One-dimensional problems.
Cauchy-Riemann equations and their applications. Analytic and Harmonic Functions. Complex integrals. Cauchy's theorem (elementary proof only), converse of Cauchy's theorem, Cauchy integral formula and its corollaries, Series - Taylor and Laurent expansion, classification of singularities, branch point and branch cut, residue theorem and evaluation of some typical integrals using this theorem.

Unit-2

Scattering Cross Section, Scattering Amplitude, Partial waves, Scattering by a central potential: Partial Wave Analysis, Significant Number of Partial Waves, Scattering by an Attractive Square –well Potential, Briet Wigner Formula, Scattering Length, Expression for Phase Shifts, Integral Equation, The Born Approximation, Scattering by Screened Coulomb Potential, Validity of Born Approximation, Laboratory and Center of Mass Coordinate systems.

Unit-3

Fluctuations: Brownian motion, Langevin theory of random motion, Time dependence of fluctuations,Power spectrum of fluctuation, Persistence and correlation of fluctuation, Wiener - Khinching theorem,Johnson noise - Nyquist theorem, Shot noise, Fokker-Planck equation.

Unit-4

Critical phenomena and phase transitions: Phase transitions, Condition for phase equilibrium, First order phase transition, Clusius– Clayperonequation, Second order phase transition, The Critical exponent, Van der Walls theory ofGas - Liquid condensation, Co - operative processes, Curie - Weiss theory of Magnetic transition,Ising Model, Ising Model in zeroth approximation, Exact solution of one dimensional Ising Model, Order parameters

Reference Books

1. Mathematical Methods for Physicists by G. Arfken Weber, Academic Press, 6th Ed (2005)
2. Mathematical Physics by P. K. Chattopadhyay (Wiley Eastern Limited, (1990)

3. Mathematical Methods for Physicists by G. Arfken Weber, Academic Press, 6th Ed (2005)

4. Modern Quantum Mechanics, J. J. Sakurai & Jim Napolitano, Pearson

5. Statistical Mechanics - Theory and Applications by S.K. Sinha, Narosa Publishing House, New Delhi.

6. Statistical Mechanics and Properties of Matter by E.S. Raja Gopal, McMillan Company of India Limited.

7. Statistical Mechanics - An Introduction by Evelyn Guha, Narosa Publishing House

8. Funadamentals of Statistical Mechanics by F. Reif, McGraw Hill Companies

9. Statistical Mechanics by R.K. SrivastavaJ.AshokmPrintice Hall of India

10. Fundamentals of Statistical Mechanics by John D. Walecka, World Scientific

11. Landau and Lifshitz, Landau theory of phase transition Statistical Physics

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M.Sc. Physics Syllabus

Semester-3

CORE Paper: Introduction to Remote Sensing and Plasma Physics
Course Code: PPH-3804
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Fundamentals of plasma physics describing variety of plasma occurring naturally/artificially
2. Single particle picture under various electromagnetic field configuration, which will help to understand concept of confinement in tokamak device.
3. Collision phenomena in plasma describing collisional transport and resistivity.
4. Fluid and kinetic description of plasma state, plasma waves and its experimental aspects
5. This overall prepare students foundation for further advanced plasma course like magnetically confined thermonuclear fusion reactor: tokamak
6. Get the end – to – end idea of remote sensing with historical perspective and current status of remote sensing
7. Understand and explain the properties of electromagnetic radiation of relevance to remote sensing
8. Understand the physical process leading to various signatures based on which targets are identified
9. Deal with sensors used for collection of remote sensing data
10. Understand the capabilities and limitation of various platforms to carry out remote sensing sensors
11. Solve problems based on above concepts.

Course Structure

Unit-1

ntroduction to Plasma Physics: Occurrence of plasma in nature, definition of plasma, concept of temperature, Debye shielding, plasma parameter, criteria for plasmas, applications of plasma physics.
Single Particle Motions: Introduction, uniform E and B fields, non-uniform B field, non-uniform E field, time-varying E field, time varying B field, adiabatic invariants

Unit-2

Plasma as Fluids: Relation of plasma physics with ordinary electromagnetics, fluid equation of motion, fluid drifts perpendicular to B, fluid drift parallel to B, plasma approximation.
Waves in Plasmas: Introduction to waves, plasma oscillations, electron plasma waves, sound waves, ion waves, validity of plasma approximation, comparison of ion and electron waves, electrostatic electron oscillations perpendicular to B, electrostatic ion waves perpendicular to B, lower hybrid frequency, electromagnetic waves with B = 0, experimental applications, electromagnetic waves perpendicular to B, cut-offs and resonances, electromagnetic waves parallel to B, hydromagnetic waves, magnetosonic waves, CMA diagram.

Unit-3

Fundamentals of Remote Sensing (RS): Definition of Remote Sensing; Advantages of Remote Sensing; Process components of Satellite Remote Sensing; Electromagnetic Radiation (EMR); Structure of Earth's atmosphere; Interaction of EMR with Earth's atmosphere and Earth's surface; Rayleigh and Mie Scattering; Absorption Blackbody radiations laws: Planck law, Rayleigh Jeans law, Wien's displacement law.

Unit-4

Ground, Air Space platforms; Polar and geostationary orbits; Types of RS sensors: Passive and Active; Examples of RS sensors; Major mechanisms of sensors: Push broom, Whisk broom; Types of Resolution: Spatial, Spectral, Radiometric, Temporal; Basics of Microwave RS: RADAR; Basics of LIDAR RS. Physical basis of spectral signatures; Basics of Digital Image Processing; Introduction to Indian RS programme; Major applications of RS.

Reference Books

1. Introduction to Plasma Physics and Controlled Fusion by Francis F. Chen
2. Fundamentals of Plasma Physics by J. A. Bittencourt
3. The Physics of Fluids and Plasmas: An Introduction for Astrophysicists by Arnab Rai Choudhuri
4. Plasma Dynamics by R. O. Dendy

5. Waves in Plasmas by Thomas H. Stix

6. Fundamentals of Remote Sensing by George Joseph, Universities Press.

7. Introductory Digital Image Processing by John R. Jensen, Pearson Publisher.

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M.Sc. Physics Syllabus

Semester-3

CORE Paper: Physics Laboratory
Course Code: PPH-3805L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to demonstrate the Frank – Hertz Experiment.
2. Will be able to d0 the Fourier Analysis.
3. Will be able to Study the Dielectrics.
4. Will be able to learn the properties and application of Lasers .
5. Will be able to demonstrate the Young's Modulus using Flexural Vibrations.
6. Will be able to demonstrate the Faraday Effect.
7. Will be able to Measure the low Resistance using Lock-in Amplifier.
8. Will be able to Measure the Permeability of Air.
9. Will understand the application of Fourier Analysis.
10. Will be ready to demonstrate application of dielectric constant and laser experiment.

Course Structure

1. Frank – Hertz Experiment
2. Fourier Analysis
3. Study of Dielectric Constant

4. Laser Experiment-II
5. Young's Modulus using Flexural Vibrations
6. Faraday Effect
7. Measurement of low Resistance using Lock-in Amplifier
8. Measurement of Permeability of Air

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M.Sc. Physics Syllabus

Semester-3

CORE Paper: Electronics Laboratory
Course Code: PPH-3806L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to demonstrate the DC Characteristic of SCR.
2. Will be able to demonstrate the SCR as Half Wave Rectifier.
3. Will be able to demonstrate the Astable Multivibrator using IC 555.
4. Will be able to demonstrate the Monostable Multivibrator using IC 555.
5. Will be able to demonstrate the UJT as Relaxation Oscillator.
6. Will be able to demonstrate the RC Phase Shift Oscillator.
7. Will be able to demonstrate the D/A Converter.
8. Will be able to demonstrate Microprocessor Interfacing with PPI 8255
9. Will understand the application of SCR and practical use.
10. Will be ready to design the circuit using SCR and Multivibrator.

Course Structure

1. DC Characteristic of SCR
2. SCR as Half Wave Rectifier
3. Astable Multivibrator using IC 555

4. Monostable Multivibrator using IC 555
5. UJT as Relaxation Oscillator
6. RC Phase Shift Oscillator
7. D/A Converter
8. Microprocessor Interfacing with PPI 8255

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M.Sc. Physics Syllabus

Semester-4

CORE Paper: Instrumentation and Microcontrollers
Course Code: PPH-4801
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Study the drawbacks of two wire transmission lines at high frequency and the application of waveguides
2. Study the microwave junction and microwave sources including the klystron family.
3. Study and work with the microcontroller 8051, and will be designing the program including data transfer operation, arithmetic, logical, Boolean, and branching programs.
4. Design the Schmitt trigger, astable multivibrator and triangular wave generator using comparator circuits.
5. Know the importance of PLL IC, demonstrate the working of phase detector, comparator and VCO.
6. Design frequency multiplicationusing PLL.
7. Explain FM demodulation, FSK demodulator and AM detection.
8. Understand different concepts of Transducers, including those for measurement of temperature, strain, motion, position and light.
9. Choose proper transducer to make sensitive measurements of physical parameters like pressure, flow, displacement, velocity, temperature etc.
10. Locate different types of transducers and sensors used in real life applications.
11. Solve numerical based on above concepts and topics

Course Structure

Unit-1

Microwaves Transmission lines and waveguides: – line equations, impedance, reflections and voltage standing wave ratio, rectangular waveguides. Scattering(S) Parameters. Microwave T-Junctions: – H-plane Tee junction, E-plane Tee- junction, E-H (Hybrid or Magic) Tee. Directional couplers, Isolator, Circulator. Microwave Sources and Devices – Two cavity Klystron amplifier, Multicavity Klystron, Reflex Klystron, Travelling wave tube (TWT), Magnetron, Gunn diode, IMPATT diode, Crystal Detector and PIN diode.

Unit-2

Microcontroller 8051 Microcontrollers-Introduction, MCS-51 Architecture, Registers in MCS- 51. A: 8051 Pin Description, Connections, I/O Ports and Memory Organization- 8051 Pin Description, 8051 Connections, 8051 Parallel I/O Ports, Memory Operations. B: Addressing Modes supported by 8051, Know the instruction set of MCS-51 microcontrollers, Data transfer instructions, Arithmetic instructions, Logical Instructions, Boolean variable manipulation instructions, program branching instructions.

Unit-3

OPAMP Applications Schmitt trigger, square wave generator ,triangle wave generator Phase Locked Loop Introduction, Basic principles, Phase Detector and comparator, voltage controlled oscillator (VCO), low pass filter, monolithic phase locked loop, PLL applications, frequency Multiplication/division, frequency translation, AM detection, FM demodulation, FSK demodulator.

Unit-4

Primary Sensing Elements and Transducers Mechanical Devices as Primary Detectors, Pressure Sensitive Primary Devices, Flow Rate Sensing Elements, Tansducers, Characteristics Choice of Transducers, Resistive Transducers, Potentiometers, Strain Gauges, Resistance Thermometers, Thermistors, Thermocouples, Variable Inductance Transducers, Capacitive Transducers, Peizo-Electric Transducers, Hall Effect Transducers, Photo voltaic cell, Semiconductor Photo Diode, Photo Transistor.

Reference Books

1. Microwave Devices Circuits, Samuel Y Liao, Pearson
2. Microwave Radar Engineering, M Kulkarni, Umesh Publications
3. Microcontrollers (Theory and Applications) By Ajay V Deshmukh.
4. Linear Integrated circuit 4th edition, By Roy Choudhury and Jain, New Age International
5. Electronic Instrumentation by H S Kalsi, TMH Edition

6. A course in Electrical Electronic Measurements Instrumentation, AK Sawhney, Dhanpat Rai Co.

7. Electronic Instrumentation and Measurement Techniques, Cooper Helfric, Prentice Hall

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M.Sc. Physics Syllabus

Semester-4

CORE Paper: Adv. Theoretical Physics and Cosmology
Course Code: PPH-4802
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. To explain the Dirac equation and its free-particle solutions, to explain the existence of antiparticles
2. Describe the classical and quantum mechanical treatment to understand many-body problem, describe the Kohn-Sham equation, pseudo-potential, exchange correlation functions for the material properties.
3. Distinguish different approaches for the density functional theory and compute its ground state energy and density, understand the modern theoretical methods based on ab initio simulations and they are so reliable that one can predict material properties.
4. To use an understanding of our galaxy to contrast and compare it with other galaxies as to type, contents, age, luminosity, motion, and size, to use cosmological models to analyse the size, age, structure, and motion of the universe overall.
5. Have a deep understanding of the physics and evolution of the smooth, homogeneous Universe starting from the Big Bang, to understand the dark matter issue and possible solutions and implications.
6. Solve numerical based on above concepts and topics

Course Structure

Unit-1

Symmetry and Conservation laws: Symmetry Transformation, Translation in Space: Conservation of Linear Momentum, Translation in Time: Conservation of Energy, Rotation in Space;

Conservation of Angular momentum, Space Inversion; Parity Conservation, Time Reversal Angular momentum General Angular Momentum, Eigen value of J2 and Jz , Angular Momentum matrices, Matrices for J+ , J , Jx and Jy, Spin angular momentum, Spin vectors for Spin –(1/2) system, Addition of Angular momenta, Clebsh- Gordan Coefficient.

Unit-2

Klein-Gordan Equation, Interpretation of the Klein-Gordon Equation, particle in a Coulomb field, Dirac's Equation for a Free Particle, Dirac Matrices, Covariant from Dirac Equation, Probability density, Plane wave solution, Negative Energy States, Spin of a Dirac Particle, Magnetic moment of the electron, Spin orbit Interaction, Radial Equation for an electron in a Central Potential, The hydrogen atom.

Unit-3

Review of molecular dynamics, Many-electron Schrodinger equation, Variational method for many-electron system, Self-consistent field method, Hartree theory and Hartree-Fock theory, Slater's treatment of HartreeFock theory, Many-electron theory in terms of the density – Thomas Fermi and Thomas Fermi-Dirac methods, Modern density-functional theory – Hohenberg-Kohn theorem and Kohn-Sham method, Treatment of exchange-correlation in density-functional theory, Approximate functionals, Applications to atoms, molecules, and solids; perturbation theory in DFT.

Unit-4

Standard Model of Cosmology, Hubble's law and the expanding universe, Big Bang theory, FRW metric, Friedman equation, different components of the universe, Distant measures in cosmology, The Cosmic Microwave Background, Recombination and Decoupling of photons, Baryon Acoustic Oscillations, Big Bang Nucleosynthesis, Inflation and the very early universe, An overview of dark matter problem.

Reference Books

1. Quantum Mechanics Leonard Schiff McGRAW-HILL PUBLISHING COMP. LTD
2. The ABC of DFT Kieron Burke and friends Department of Chemistry, University of California, Irvine.
3. Electronic Structure: Basic Theory and Practical Methods by R. Martin
4. Introduction to Cosmology, Barbara Ryden, Cambridge University Press; 2 edition
5. An introduction to modern cosmology, Andrew Liddle, Wiley-Blackwell; 2nd edition
6. Modern cosmology, Scott Dodelson, Elsevier

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M.Sc. Physics Syllabus

Semester-4

CORE Paper: Adv. Nuclear Physics
Course Code: PPH-4803
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Understand the importance of models in describing the properties of nuclei and nuclear collisions.
2. Understand how various types of accelerators work and understand differences between them.
3. Understand major applications of accelerators and the recent new concepts.
4. Able to account for central concepts within particle physics, such as symmetries, invariants, and conservation laws.
5. Understanding of nuclear reactions mechanisms, compound nuclei and direct reactions.
6. Able to understand radioactive ion beams and applications and understand of nucleosynthesis.
7. Understand how fusion process in the sun and stars take and understanding of controlled thermonuclear fusion.
8. Able to understand the newer concepts in particle acceleration and Laser Wake-field acceleration.
9. Understanding the compact particle accelerators for application in medical sciences and health-care wellness.
10. Understand of Parity non-conservation in weak-interaction and relativistic kinematics.
11. Solve numerical based on above concepts and topics.

Course Structure

Unit-1

Nature of nuclear force, form of nucleon-nucleon potential, charge-independence and charge-symmetry of nuclear forces, Deuteron problem. Evidence of shell structure, single particle shell model, its validity and limitations, Rotational spectra.

Unit-2

Nuclear Reactions mechanisms, compound nuclei and direct reactions, concept of surrogate technique for measurement of reaction cross section on radionuclide targets. Radioactive Ion Beams and applications, Nucleosynthesis, Accelerator Driven Systems, Advances in 4th Gen Fission Reactors.

Unit-3

Fusion Process in the Sun and Stars, Controlled Thermonuclear Fusion, Confinement schemes, elementary Fusion Reactor design concepts, Fusion Neutronics, Detection of fast neutrons(diamond detectors– natural and CVD-diamond), concept of Fission-Fusion Hybrid Reactors.

Unit-4

Newer concepts in particle acceleration, plasma acceleration, Laser Wake-field acceleration, compact particle accelerators for application in Medical Sciences, Health-care Wellness, Medical Diagnosis and Radiation Therapy. C.P. and T' invariance, Application of symmetry arguments to particle reactions, Parity non-conservation in weak-interaction, relativistic kinematics.

Reference Books

1. Introductory Nuclear Physics, Kenneth S. Krane, Wiley
2. An Introductory Course of Particle Physics, Palash Baran Pal, CRC Press

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M.Sc. Physics Syllabus

Semester-4

CORE Paper: Atmospheric Sciences and Astronomy
Course Code: PPH-4804
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Understand the relation between astronomy and astrophysics.
2. Apply principles of physics to astronomical objects.
3. Introduce students to the field of astrophysics with mathematically based principles.
4. Understanding the Interstellar medium and star formation.
5. Able to understand the telescopes and its different types and the resolution.
6. Understanding the image formation in a camera and focal plane instruments.
7. Able to understand the photometers, spectrometers and detectors.
8. Get the knowledge of radiative transfer within different layers of Atmosphere
9. Able to solve problems related to atmospheric science using their computational skills
10. Understand the vast field of Atmospheric science
11. Solve numerical based on above concepts and topics

Course Structure

Unit-1

Celestial Sphere; Coordinate systems; Time; Observable quantities; Continuum radiation from Stars; Terminology - Brightness, Luminosity, Magnitude scale, colour; Size and Distance; Stellar

spectra and spectral classification of stars. Formation of spectral lines - line broadening - curve of growth; Local Thermodynamic Equilibrium - Saha's equation; HR diagram; Binary stars and determination of stellar parameters.

Unit-2

Main sequence phase of stars - Energy sources; Equations for Stellar interiors - stability; Atmospheres of stars; Post-main sequence evolution of stars; Fate of stars at the late stages of evolution - Mass loss - Planetary nebulae - supernovae; Chandrasekhar limit - Degenerate core remnants - White Dwarfs - Neutron stars - Black Holes; Interstellar medium and Star formation; Galaxies and their classification; Hubble's law; Introduction to Active Galactic Nuclei and Gamma Ray Bursts; Extra-solar planets.

Unit-3

A: Telescopes - Different types of telescopes - Angular resolution and Diffraction Limited Resolution - Image formation in a camera - Plate Scale - Observatories (Ground Based Space Based) - Focal Plane Instruments: Imagers - Photometers - Spectrometers – CCDs and their use in astronomy - Detectors for Optical, Infrared, UV, X-rays, and Gamma-rays - Effect of Atmosphere (Seeing and Scintillation)
B: Observational cosmology

Unit-4

Fundamentals of Radiative Transfer; Radiative transfer equation and solution of it in simple cases; Remote Sensing; Radiative Effects of aerosols.

Reference Books

1. Fundamental Astronomy, Hannu Karttunen, Springer
2. ||Radiative Transfer in the Earth System|| by Charlie Zender

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M.Sc. Physics Syllabus

Semester-4

CORE Paper: Physics Laboratory
Course Code: PPH-4805L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be able to demonstrate the Zeeman Effect.
2. Will be able to demonstrate Hysteresis Loop Tracer.
3. Will be able to demonstrate the Electron Spin Resonance (ESR)
4. Will be able to demonstrate the Optical Fibre Experiment.
5. Will be able to measure the Permittivity of Air.
6. Will be able to demonstrate the Kerr Effect.
7. Will be able to measure Mutual Inductance using Lock-in Amplifier.
8. Will be able to demonstrate Phase Locked Loop (PLL).
9. Will be able to demonstrate Data Acquisition using Coupled Pendulum and Microcontroller 8081.
10. Will be able to demonstrate Pulse Width Modulation and LVDT Transducer and Strain Gauge

Course Structure

1. Zeeman Effect
2. Hysterisis Loop Tracer

3. Electron Spin Resonance(ESR)
4. Optical Fibre Experiment
5. Measurement of Permittivity of Air
6. Kerr Effect
7. Measurement of Mutual Inductance using Lock-in Amplifier
8. Phase Locked Loop(PLL)
9. Data Acquisition using Coupled Pendulum
10. Microcontroller 8081
11. Pulse Width Modulation
12. LVDT Transducer and Strain

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M.Sc. Physics Syllabus

Semester-4

CORE Paper: Project and Dissertation
Course Code: PPH-4806L
No. of credits: 4
Learning Hours: 60

Course Objectives

At the end of the course student will be able to

1. Will be trained in identifying Projects by doing literature survey in forms of Research papers, journals and looking for ideas in internet.
2. He/She is also encouraged to come with original ideas which explain the concepts of Physics and electronics.
3. Will be trained in having —Hands on experience|| with designing projects using various instruments, collecting data and in its analyses.
4. Will be able to document his project by writing synopsis and project report

Course Structure

Students are expected to finish their Project work by the end of Semester 4 and submit their Dissertation. The internal as well as external evaluation will be based on their Project.